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DEPARTMENT VISION AND MISSION

a. Vision Statement:

To be a centre of excellence in the realm of Commerce and Management, developing and nurturing global competencies in students through quality education, research and continuous innovation.

b. Mission Statement:

To nurture commerce professionals who possess a high level of knowledge and competence to effectively contribute to society with commitment and integrity.

INTRODUCTION TO BACHELOR OF COMMERCE (WORK INTEGRATED)

PROGRAMME OUTCOME

Reference No	Statement	Attribute
PO1	Demonstrate updated knowledge and understanding of key theories and practices in the core areas of Commerce.	Academic Excellence: Domain Expertise, Knowledge Application
PO2	Exhibit professional skills including communication, teamwork and problem-solving required for effective performance.	Academic Excellence: Problem Solving Skills Communication: Verbal Skills Leadership: Team Building Skills
PO3	Develop an appreciation of analytics in the form of data-driven metrics to business activities and how these can be used.	Academic Excellence: Knowledge Application, Analytical Skills, Problem Solving Skills
PO4	Facilitate the development of skills that are directly relevant to the contemporary workplace and future of work.	Professional Excellence: Employability, Practical Skills Leadership: Dynamism
PO5	Exhibit ethical awareness and professional integrity in business practices, promoting inclusivity and sustainability.	Leadership: Inclusiveness Personality: Personal Integrity Social Sensitivity: Respecting Diversity, Sustainability

COURSE STRUCTURE

Bachelor of Commerce (Work Integrated)

SEMESTER – I

Course Code	Course Name	Course Type	Category	Total Marks	L-Hr/ Week	T-Hr/ Week	P-Hr/ Week	Credits
ENG181-1	Communicative English	Theory	Ability Enhancement Course	50.00	2			2
COM101-1	Business Laws	Theory	Major (Core)	100.00	4			4
CWI101-1	Financial Accounting	Theory	Major (Core)	100.00	4			4
CWI161-1	Business Intelligence with Spreadsheets	Theory	Skill Enhancement Course	100.00	4			4
-	Multi-Disciplinary Course	-	Multi-Disciplinary Course	100.00	3			3
CWI181-1	Indian Accounting (IND AS)	Theory	Value Added Course	50.00	2			2
CWI182-1	Indian Knowledge System	Theory	Value Added Course	50.00	2			2
EVS181-1	Environmental Studies	Theory	Value Added Course	50.00	1			1
HED181-1	Holistic Education	Theory	Value Added Course	50.00	1			1

Bachelor of Commerce (Work Integrated)

SEMESTER – II

Course Code	Course Name	Course Type	Category	Total Marks	L-Hr/ Week	T-Hr/ Week	P-Hr/ Week	Credits
ENG181-2	Communicative English II	Theory	Ability Enhancement Course	50.00	2			2
COM102-2	Business Economics	Theory	Major (Core)	100.00	4			4
CWI101-2	Statistics for Management	Theory	Major (Core)	100.00	4			4
CWI102-2	Foundations of Financial Markets	Theory	Major (Core)	100.00	4			4
-	Multi-Disciplinary Course		Multi-Disciplinary Course	100.00	3			3
CWI181-2	Data Analytics using Python	Theory	Value Added Course	50.00	2			2
HED181-2	Holistic Education	Theory	Value Added Course	50.00	1			1
UOI181-2	Understanding of India	Theory	Value Added Course	50.00	1			1

Bachelor of Commerce (Work Integrated)

SEMESTER – III

Course Code	Course Name	Course Type	Category	Total Marks	L-Hr/ Week	T-Hr/ Week	P-Hr/ Week	Credits
-	Language	Theory	Ability Enhancement Course	50.00	2			2
CWI201-3	Corporate Accounting	Theory	Major (Core)	100.00	4			4
CWI202-3	Strategic Human Resource Management	Theory	Major (Core)	100.00	4			4
CWI203-3	Data Visualization and Dashboarding	Theory	Major (Core)	100.00	4			4
-	SWAYAM Course	-	Multi-Disciplinary Course	100.00	3			3
CWI281-3	Investment Banking	Theory	Value Added Course	50.00	2			2
HED181-3	Holistic Education	Theory	Value Added Course	50.00	1			1
CWI461-3	Work Integrated Learning Practicum	Practical	Skill Enhancement Course	50.00	2			2

Bachelor of Commerce (Work Integrated)

SEMESTER – IV

Course Code	Course Name	Course Type	Category	Total Marks	L-Hr/ Week	T-Hr/ Week	P-Hr/ Week	Credits
-	Language	Theory	Ability Enhancement Course	50.00	2			2
CWI201-4	Financial Management	Theory	Major (Core)	100.00	4			4
COM201-4	Entrepreneurship	Theory	Major (Core)	100.00	4			4
CWI202-4	Modern Banking Practices	Theory	Major (Core)	100.00	4			4
CWI261-4	Digital Marketing and AI	Practical	Skill Enhancement Course	100.00	3			3
CWI461-4	Work Integrated Learning Practicum	Theory	Skill Enhancement Course	100.00	4			4
CWI281-4	Financial Modelling and Valuation	Theory	Value Added Course	50.00	2			2
HED181-4	Holistic Education	Theory	Value Added Course	50.00	1			1

Bachelor of Commerce (Work Integrated)

SEMESTER – V

Course Code	Course Name	Course Type	Category	Total Marks	L-Hr/ Week	T-Hr/ Week	P-Hr/ Week	Credits
CWI201-5	Direct Taxation Law	Theory	Major (Core)	100.00	4			4
CWI301-5	Cost & Management Accounting	Theory	Major (Core)	100.00	4			4
COM101-5	Research Methodology	Theory	Major (Core)	100.00	4			4
CWI302-5	Financial Statement Analysis	Theory	Major (Core)	100.00	4			4
CWI461-5	Work Integrated Learning Practicum	Theory	Skill Enhancement Course	100.00	4			4
CWI281-5	Generative AI for Business	Theory	Value Added Course	50.00	2			2

Bachelor of Commerce (Work Integrated)

SEMESTER – VI

Course Code	Course Name	Course Type	Category	Total Marks	L-Hr/ Week	T-Hr/ Week	P-Hr/ Week	Credits
CWI301-6	Workplace Behaviour and Dynamics	Theory	Major (Core)	100.00	4			4
CWI302-6	Auditing and Assurance	Theory	Major (Core)	100.00	4			4
CWI303-6	International Business	Theory	Major (Core)	100.00	4			4
CWI461-6	Work Integrated Learning Practicum	Practical	Skill Enhancement Course	100.00	4			4
CWI361-6	GST Filing: Practice and Procedures	Practical	Skill Enhancement Course	50.00	2			2
CWI281-6	ESG and Corporate Sustainability	Theory	Value Added Course	50.00	2			2

SEMESTER – I SYLLABUS FOR

COURSE NAME: BUSINESS LAWS

COURSE CODE: COM101-1

COURSE TYPE: Theory

TOTAL HRS: 4 Hours/Week

Course learning Outcomes

CO1: Understand the basic rules of Agreements and Contracts along with the rules of Offer, Acceptance, Consideration, Capacity/Competency to contract; the rules governing in The Indian Contract Act 1872.

CO2: Analyze the concept of free consent, analyse its implications in a contract, examine the legality of the object, and interpret key judicial decisions under the Indian Contract Act.

CO3: Comprehend the performance and discharge of contracts, contracts of indemnity, guarantee, bailment, pledge, lien, and agency, including their essentials, types, and legal implications.

CO4: Discern the principles of the contract of sale, including its essentials, conditions and warranties, transfer of ownership, rights of an unpaid seller, performance obligations, and the doctrine of caveat emptor.

CO5: Recognize the fundamentals of competition, identify fair and unfair competitive practices, analyze dominant positions and malicious economic practices, examine anti-competitive agreements and combinations, and evaluate the role and functioning of the Competition Commission of India (CCI).

Course Content:

U1: The Indian Contract Act, 1872- General Contracts-I

Teaching Hours: (15)

Contract – meaning, characteristics, and kinds; Essentials of valid contract – Offer- meaning, Essential elements of a valid offer, types of offer; Acceptance- meaning and valid rules of acceptance; Communication of offer and acceptance, Revocation of offer and acceptance, Lapse of Offer, Consideration- meaning, essential elements, legal rules regarding consideration, exceptions to the general rule of consideration, Rule of Privity of contract and its exceptions; Contractual Capacity- meaning and types of parties, persons not competent to contract, Nature and effects of Minor's agreements. Landmark judgments. (Meaning includes definitions given by the Act)

U2: The Indian Contract Act, 1872- General Contracts-II

Teaching Hours: (15)

Free Consent- meaning and Five elements which affect the consent and the legal rules of each of the elements (essentials, effects, and remedies); Legality of Objects- meaning, agreements opposed to public policy; illegal agreements, their effects; Void Agreements- meaning and contracts expressly declared void (types of void agreements); Contingent contracts- meaning and essentials, Quasi contracts- meaning; Landmark judgments. (Meaning includes definitions given by the Act).

U3: The Indian Contract Act, 1872- General & Specific Contracts Teaching Hours: (10)

Performance of contracts- meaning and types, Discharge of contracts – modes of discharge including breach and its remedies. Concept and essentials of Indemnity and Guarantee, Contract of Bailment-essentials and Kinds; Lien and its types; Pledge- meaning and essentials; Contract of Agency -Creation of Agency – Kinds of Agents –Termination of Agency. Landmark judgments. (Meaning includes definitions given by the Act).

U4: The Sale of Goods Act, 1930

Teaching Hours: (10)

Contract of sale - meaning, essentials, and the difference between sale and agreement to sell; Conditions and warranties- meaning, essentials and differences; Transfer of ownership in goods including sale by non-owners, Performance of the contract of sale, Unpaid seller – meaning and rights of an unpaid seller against the goods and the buyer. The Doctrine of Caveat Emptor: Landmark Judgments.

U5: Competition Act, 2002

Teaching Hours: (10)

Competition- Meaning and Nature, Fair and Unfair competitive practices, Dominant Position, Malicious Economic Practices, Anti-Competitive Agreements, Types and sub-types of Anti-Competitive Agreements, Definition of Combination, Competition Commission of India (CCI), Establishment Composition. Landmark judgments.

Essential Reading:

Arora, S. (2025). *Business Laws*. Taxmann's Publications. ISBN 978-9371263931

Garg, K. C., Sareen, V. K., Sharma, M., & Sharma, G. (2022). *Business Laws*. Kalyani Publishers. ISBN 978-9355400741

COURSE NAME: FINANCIAL ACCOUNTING

COURSE CODE: CWI101-1

COURSE TYPE: Theory

TOTAL HRS: 4 Hours/Week

Course learning Outcomes

CO1: Compare the accounting system adopted in hire purchase and instalment system.

CO2: Estimate royalty under different situations.

CO3: Ascertain the profit earned on consignment.

CO4: Analyse the branch transactions to calculate profit or loss

Course Content:

U1: Accounting for Hire-Purchase and Instalment Systems Teaching Hours: (15)

Hire Purchase - Meaning - Legal provisions, [including repossession – only theory] - Calculation of interest - when the rate of interest and cash price is given - when cash price and the total amount payable is given when the rate of interest and instalments amount is given, but cash price is not given - Calculation of cash price under annuity method - Journal entries and Ledger accounts in the books of hire-vendor. Instalment system- Meaning - Difference between hire purchase and instalment system.

U2: Royalty Accounts Teaching Hours: (15)

Introduction - meaning - technical terms - royalty - the landlord-tenant - minimum rent - short workings - recoupment of short working under - fixed period - floating period - recoupment within the life of a lease - treatment of strike, stoppage of work and sub-lease - accounting treatment in the books of the lessee(tenant) - when royalty is less than minimum rent - when royalty is equal to minimum rent - when the right of recoupment is lost - when minimum rent account method is followed - preparation of ledger accounts - royalty account - landlord account – short workings account - minimum rent account when minimum rent account is followed.

U3: Consignment Teaching Hours: (15)

Consignment: Features – Del-credere commission - Accounting treatment in the books of the consignor and consignee – Valuation of unsold stock – Accounting treatment of Normal loss

and abnormal loss – Abnormal loss account - Goods sent to consignee at invoice price –
Accounting treatment – stock reserve calculation.

U4: Accounting for Inland Branches

Teaching Hours: (15)

Concept of dependent branches; accounting aspects; debtors' system – cost price method –
Invoice price method – Branch Debtors account (problems)—Independent branches: concept-
accounting treatment: adjustment entries.

Essential Reading:

Gupta, R. L., & Radhaswamy, M. (2024). *Financial Accounting*. New Delhi: Sultan Chand & Sons.

Jain & Narang. (2024). *Advanced Accountancy*. Mumbai: Kalyani Publishers.

COURSE NAME: BUSINESS INTELLIGENCE USING SPREADSHEETS

COURSE CODE: CWI161-1

COURSE TYPE: Practical

TOTAL HRS: 4 Hours/Week

Course learning Outcomes

CO1: Demonstrate basic data cleaning and pre-processing in Excel.

CO2: Apply Financial, Logical, Statistical, Reference and Text Excel functions to solve problems.

CO3: Construct payroll, ratio analysis, cash flow statement of a business using Excel VBA.

CO4: Design and build interactive dashboards and reports using charts, pivot tables, and advanced spreadsheet functions.

Course Content:

U1: Introduction to Business Intelligence

Teaching Hours: (08)

Concept of Business Intelligence - Role of BI in modern organizations; BI vs Business Analytics vs Data Analytics; Types of business data (structured, semi-structured, unstructured); Importing data from multiple sources (CSV, text files, databases, web); Handling missing, duplicate, and inconsistent data; Data normalization concepts - BI lifecycle and decision-making process.

U2: Basic Data Processing

Teaching Hours: (10)

Spreadsheet - Workbook - Cell Referencing, Cell Addressing, File Menu; Home Menu, Range, Conditional Formatting, formatting as a Table, Cell Styles, AutoSum, Sort and Filter; Insert Menu: Inserting Tables and Pivot Tables, Smart Arts, Charts; Page Layout, Review and View Menus; Freeze panel Data view: Converting Text to Columns, Removing Duplicates, Data Validation, Grouping and Ungrouping. Basic Excel functions: Flash fill, Sum, Average, Min, Max, Count.

U3: Financial, Logical and Text Functions

Teaching Hours: (12)

Text Functions: UPPER, LOWER, LEFT, RIGHT, TRIM, T, TEXT, LEN, DOLLAR, EXACT, CONCATENATE; Logical Functions: AND, OR, NOT, IF, TRUE; Financial

Functions: Depreciation (DB, DDB, VDB), Simple Interest (PMT, NPER, INTRATE) - Present Value, Net Present Value, Future Value (PV, NPV, FV) - Internal Rate of Return (IRR, MIRR); Practical exercises based on Text, Logical and Financial functions.

U4: Statistical and Reference Functions

Teaching Hours: (12)

Date & Time Functions: DATE, DATEVALUE, DAYS360, NOW, TIME, TIMEVALUE, WORKDAY, WEEKDAY, YEAR; Lookup and Reference Functions: HLOOKUP, VLOOKUP, TRANSPOSE, GETPIVOTDATA, HYPERLINK; Statistical Functions: Mean, Median, Mode, Standard Deviation, Correlation, Skewness, Chi-Square Analysis; Practical Exercises.

U5: Applications of Excel in Business:

Teaching Hours: (10)

Ratio Analysis, Cash Flow Statement, Payroll Processing, Marketing, Sales and Advertising Data Analytics, Social Media Marketing Analysis, Basic applications with Macros and VBAs; Trending business applications using MS Excel.

U6: Capstone Project

Teaching Hours: (08)

End-to-end BI project using spreadsheets; Problem definition and dataset selection; Data cleaning, analysis, and visualization; Final dashboard/report presentation.

References:

Winston, W. (2025). *Microsoft Excel Data Analysis and Business Modeling*, 7th Edition. Pearson Publishing.

Sharda, R. (2023). *Business Intelligence, Analytics, Data Science, and AI*, 5th Edition. Pearson Publishing.

COURSE NAME: Indian Accounting Standards (Ind AS)

COURSE CODE: CWI181-1

COURSE TYPE: Theory

TOTAL HOURS: 2 Hours/Week

Course Learning Outcomes

CO1: Understand the need and framework of Indian Accounting Standards (Ind AS).

CO2: Explain the conceptual framework and elements of financial statements under Ind AS.

CO3: Describe the basic accounting treatment under major Ind AS at a conceptual level.

CO4: Interpret Ind AS-compliant financial statements and key disclosures.

CO5: Compare Indian GAAP with Ind AS and understand the impact of transition.

Course Outline:

U1: Introduction to Ind AS and Conceptual Framework Teaching Hours: (6)

Introduction to accounting standards in India – Need for accounting standards – Limitations of traditional accounting practices – Evolution from Indian GAAP to Ind AS – Convergence with IFRS – Applicability and roadmap of Ind AS in India – Structure and framework of Ind AS – Conceptual Framework for Financial Reporting – Objectives of financial reporting – Qualitative characteristics of financial information – Elements of financial statements – Recognition and measurement principles – Overview of differences between Indian GAAP and Ind AS.

U2: Ind AS on Presentation of Financial Statements Teaching Hours: (6)

Ind AS 1 – Presentation of Financial Statements – Components of financial statements – Statement of Financial Position – Statement of Profit and Loss – Statement of Changes in Equity – Notes to accounts – Concept of materiality – Ind AS 7 – Statement of Cash Flows – Classification of cash flows into operating, investing and financing activities – Direct and indirect methods – Ind AS 8 – Accounting Policies, Changes in Accounting Estimates and Errors – Ind AS 10 – Events after the Reporting Period.

U3: Ind AS on Accounting for Assets Teaching Hours: (6)

Ind AS 16 – Property, Plant and Equipment – Recognition – Initial and subsequent measurement – Depreciation and useful life – Component accounting – Ind AS 38 – Intangible Assets – Recognition and measurement – Ind AS 40 – Investment Property – Basic

concepts – Ind AS 36 – Impairment of Assets – Indicators of impairment and accounting treatment.

U4: Ind AS on Liabilities, Provisions and Revenue

Teaching Hours: (6)

Ind AS 37 – Provisions, Contingent Liabilities and Contingent Assets – Recognition and measurement – Ind AS 12 – Income Taxes – Current tax and deferred tax – Ind AS 115 – Revenue from Contracts with Customers – Five-step model for revenue recognition – Performance obligations – Timing of revenue recognition – Disclosure requirements.

U5: First-time Adoption of Ind AS and Comparative Analysis Teaching Hours: (6)

Ind AS 101 – First-time Adoption of Indian Accounting Standards – Opening Ind AS balance sheet – Exemptions and exceptions – Impact of transition on financial statements – Comparison of Indian GAAP and Ind AS – Broad comparison of Ind AS and IFRS – Challenges in implementation of Ind AS – Recent developments in Ind AS.

Essential Reading:

Gupta, R. L., & Radhaswamy, M. *Financial Accounting*. Sultan Chand & Sons.
Jain & Narang. *Advanced Accountancy*. Kalyani Publishers.
ICAI. *Indian Accounting Standards (Ind AS)* – Latest Edition.

COURSE NAME: Indian Knowledge System for Business and Society

COURSE CODE: CWI181-1

COURSE TYPE: Theory

TOTAL HRS: 2 Hours/Week

Course Outcomes:

CO1: Describe the historical evolution and characteristics of Indian knowledge traditions.

CO2: Apply traditional approaches to risk management, negotiation, and conflict resolution in organizational contexts.

CO3: Analyze Indian perspectives on sustainability and harmony with nature

CO4: Evaluate the application of IKS in addressing contemporary societal and global challenges.

Course Content:

U1: Introduction to Indian Knowledge Systems

Teaching Hours: (06)

Scope of Indian Knowledge Systems (IKS); Characteristics of knowledge traditions in India; Historical and cultural contexts of Indian Knowledge System; Oral and written transmission of knowledge; Interdisciplinary nature of IKS; Relationship between philosophy, science, art, and society.

U2: IKS for Leadership and Communication

Teaching Hours: (08)

Insights from Arthashastra – Leadership: Concept of Rajarshi, Leadership competencies; Risk Management: Identification of external risks, crisis preparedness, and contingency planning; Negotiation: Principle of persuasion, negotiation strategies, managing conflicts through diplomacy; Panchatantra as managerial wisdom: Lessons on leadership and communication.

U3: Health, Ecology, and Sustainable Living

Teaching Hours: (08)

Ayurveda: principles and practices; Yoga and holistic well-being; Traditional ecological knowledge and biodiversity conservation; Indian perspectives on sustainability and harmony with nature; Food systems, agriculture, and indigenous practices.

U4: Contemporary Relevance of IKS

Teaching Hours: (08)

IKS and Sustainable Development: Traditional ecological knowledge and climate resilience, Indigenous resource management practices; IKS and Innovation Ecosystems: Traditional technologies and modern innovations, Indigenous knowledge as a driver of entrepreneurship; Addressing current societal and global challenges through IKS; Case studies and discussions.

Essential Reading:

1. Mahadevan, B., Bhat, B.N. and Vinayak Rajat, Nagendra Pavana R.N. (2022). *Introduction to Indian Knowledge Systems: Concepts and Applications*. PHI Learning.
2. Bhatt, V., and Shukla, T. (2024). *Kautilya's Arthashastra: Timeless Strategies for Modern Governance*. Ministry of Education, Government of India.
3. Radhakrishnan, S. (2009). *Indian Philosophy*. Oxford University Press.
4. Kautilya. (2000). *The Arthashastra*. L. N. Rangarajan (Ed.). Penguin UK.

SEMESTER – II SYLLABUS FOR

COURSE NAME: BUSINESS ECONOMICS

COURSE CODE: COM102-2

COURSE TYPE: Theory

TOTAL HRS: 4 Hours/Week

Course Learning Outcomes

CO1: Develop the conceptual foundations and analytical methods used in micro economics.

CO2: Develop the ability to understand and appreciate the economic theories and their application in real economic life.

CO3: Understand the role of prices in allocating scarce resources in market economies and explain the consequences of government policies in the form of price controls.

CO4: Appraise the monetary policy and fiscal policy prevalent within a country.

Course Content:

U1: Demand and supply Analysis and Applications

Teaching Hours: (12)

Economic models- Production Possibility Frontier and Circular Flow of Income: Two Sector Model; Demand Analysis: Law of demand, Exceptions to the law; Changes in demand, Elasticity of Demand: Definition, degrees and measurement - Supply Analysis: Laws of supply, Changes in supply, and elasticity of supply - Market equilibrium, Applications in real life: Price Ceiling, Price Floor and Externality.

U2: Theory of Consumer Choice

Teaching Hours: (08)

The Budget Constraint: What the Consumer Can Afford, Preferences: What the Consumer Wants, Indifference curves- Properties, Optimization: What the Consumer Chooses and definition of Price Effect, Income Effect and Substitution Effect.

U3: Theory of Production and Cost

Teaching Hours: (10)

Production function; Law of variable proportions; Laws of returns, Economies of scale; Cost Function-Important cost concepts. Short run and long run cost analysis (traditional theory) Modern theory of cost- Revenue analysis. TR, AR and MR.

U4: Market Structure

Teaching Hours: (15)

Market structure- Perfect competition, Price and output determination- Monopoly- Price output determination, Price discrimination Monopolistic Competition – Features, Price and Output determination (Short Run and Long Run). Oligopoly: Definition and types of Oligopoly.

U5: Measuring Nation's Income and Cost of Living

Teaching Hours: (09)

Measuring GDP - Real and Nominal GDP and GDP Deflator – Inflation: Consumer Price Index, Comparison of CPI and GDP Deflator and Adjusting economic variables to inflation – Money Supply Measures.

U6: Money and the Financial System

Teaching Hours: (06)

Saving and Investment in National Income Accounts. The Market for Loanable Funds. Banks and the Money Supply. Monetary Measures. Meaning of Inflation and Costs of Inflation. Tools of Credit Control.

Essential Reading

Lipsey, R.G. and K.A. Chrystal (2020). *Principles of Economics*. Oxford University Press.

Pindyck, R.S. and Rubinfeld, D.L. (2021). *Micro Economics*. Pearson Publishing.

COURSE NAME: STATISTICS FOR MANAGEMENT

COURSE CODE: CWI101-2

COURSE TYPE: Practical

TOTAL HRS: 4 Hours/Week

Course Learning Outcomes:

CO1: Understand basic statistical concepts and functions used for data analysis.

CO2: Build comprehensive datasets using internet platforms like Google Forms.

CO3: Apply descriptive and inferential statistical techniques in Excel to analyze real-world datasets.

CO4: Present statistical results obtained in Excel to support data-driven decisions.

Course Content:

U1: Introduction to Statistics

Teaching Hours: (10)

Origin and evolution of statistics;

Statistics in decision-making;

Sources of data - Primary data and Secondary data;

Types of Data - Qualitative and Quantitative data, Discrete and Continuous data;

Main Methods of Data Collection - Surveys and Interviews,

Data Instruments – Questionnaires and Schedules; Data Collection Exercises.

U2: Sampling, Estimation and Hypothesis testing:

Teaching Hours: (10)

Sampling – Probabilistic and Non-Probabilistic techniques;

Statistic, Parameter, Point and Interval Estimates, Confidence intervals, Level of Significance;

Hypothesis Testing: Types; How to formulate a hypothesis; Process of hypothesis testing.

U3: Descriptive Statistics

Teaching Hours: (15)

Measures of Central Tendency: Mean, Median, Mode

Measures of Variation: Range, Standard Deviation, Quartile Deviation

Measures of Shape: Kurtosis, Skewness, Histogram

Measures of Position: Partition Values, Five Number Summary.

U4: Inferential Statistics

Teaching Hours: (15)

Mean Differences: Dependent t test, Independent t test, and ANOVA;

Correlation: Pearson's correlation coefficient, Spearman's Rank correlation, Probable Error.

Regression: Meaning and applications; Linear Regression – Regression Coefficients;

Regression lines – X on Y, Y on X; Writing Regression Equations.

U5: Time-Series Analysis

Teaching Hours: (10)

Time Series – Meaning and Components; Time Series Data Vs Longitudinal Data Vs Panel Data, Additive and multiplicative models; Correlogram and its applications, Trend analysis: Finding trend by semi-average method, moving average method and least squares method; Problems on all concepts using Excel.

Essential Reading:

S.P.Gupta, *Statistical Methods*. Sultan Chand and Sons.

Das and Swain, *Business Statistics for Managerial Decision*, Himalaya Publishing House

Levin, R. I., & Rubin, D. S. (2017). *Statistics for Management*. Pearson.

Anderson, D. R., Sweeney, D. J., Williams, T. A., Freeman, J., & Shoemith, E. (Latest Ed). *Statistics for Business and Economics*. Cengage Learning, Inc.

COURSE NAME: FOUNDATIONS OF FINANCIAL MARKETS

COURSE CODE: CWI102-2

COURSE TYPE: Theory

TOTAL HRS: 4 Hours/Week

Course Learning Outcomes

CO1: Describe and classify the major components of the Indian financial system and explain their interrelationships.

CO2: Analyse and compare money market and capital market instruments, trading mechanisms, and the regulatory provisions surrounding them.

CO3: Apply and evaluate concepts of financial services to distinguish fund-based and fee-based services and assess their relevance in financing decisions.

CO4: Evaluate emerging technology-driven financial services and their applications and implications for the future of the financial services sector.

Course Content:

U1: Indian Financial System

Teaching Hours: (12)

Structure of Indian Financial System - Financial Institutions, Financial Services, Financial Markets, Financial Assets, Financial Intermediaries, Financial Instruments and Financial Regulators; Classification and Components of Financial Markets; Formal and Informal Financial Sectors, New Financial Instruments; Key Elements of well-functioning financial system, Financial Sector Reforms in India.

U2: Money Market

Teaching Hours: (12)

Trading in the Money Market; Difference between Money Market and Capital market; Money Market Instruments – Treasury Bills, Certificate of Deposits, Commercial Paper, Call Money, Notice Money, Inter Bank Participation Certificate; Structure of the Indian Money Market; Deficiencies in the Indian Money Market; RBI's and Commercial Banks' role in Money Market; Recent Trends in the Money market.

U3: Capital Market

Teaching Hours: (12)

Classification of capital markets; Stock exchanges - Concept, Functions, Listing Requirements, Major Stock Exchanges in India and abroad - BSE, NSE, NYSE, NASDAQ, LSE and OTCEI; Initial Public Offerings (IPO) – Different Kinds of Issues and related requirements; Depositories and their increasing role; SEBI as capital market regulator - Objectives, Powers, Organization, SEBI guidelines on primary markets, secondary markets; Derivatives in India - Introduction to Financial and Commodity Derivatives.

U4: Introduction to Financial Services

Teaching Hours: (14)

Features of financial services; Classification of Financial Services – Fund based and fee based; Fund-Based Services – Equipment financing, Hire Purchase, Factoring, Forfaiting, Housing Finance, Mutual Funds, Venture Capital; Fee-Based Services – Merchant Banking, Issue Management, Credit Rating, Portfolio Management, Advisory Services, Loan Syndicate, Mergers and Acquisitions, Stock broking, Underwriting.

U5: Emerging technologies in finance services

Teaching Hours: (10)

New Age Financial Services - Fintech, Regtech, Insuretech, Crowdfunding, P2P Lending, Robo Advising; Regulatory Sandbox of RBI, Applications of AI in Financial Services, Challenges facing the Financial Services sector – Present Scenario.

Essential Reading

Khan, M.Y. (2019). *Indian Financial System*. Tata McGraw Hill Publishing.

Gordon & Natarajan. (2022), *Financial Markets and Services*. Himalaya Publishing House.

Parameswaran, S. K. (2022). *Fundamentals of financial instruments: An introduction to stocks, bonds, foreign exchange, and derivatives*. John Wiley & Sons.

COURSE NAME: Data Analytics using Python

COURSE CODE: CWI181-2

COURSE TYPE: Practical

TOTAL HOURS: 2 Hours/ Week

Course Learning Outcomes

CO1: Understand the fundamentals of data analytics and the role of Python in data analysis.

CO2: Build, import, and manage datasets using Python libraries.

CO3: Apply descriptive and basic inferential statistical techniques using Python.

CO4: Visualize data and present analytical results to support data-driven decision-making.

Course Outline:

U1: Introduction to Data Analytics and Python

Teaching Hours: (6)

Introduction to data analytics – Role of data analytics in business and decision-making – Types of analytics: descriptive, diagnostic, predictive, prescriptive – Introduction to Python – Features and applications of Python – Python environment setup – Jupyter Notebook and Anaconda – Basic Python syntax – Variables and data types – Input and output operations.

U2: Python Programming for Data Analysis

Teaching Hours: (6)

Python operators and expressions – Conditional statements – Looping structures – Functions – Introduction to NumPy – Arrays and array operations – Introduction to Pandas – Series and Data Frames – Importing datasets (CSV, Excel) – Basic data inspection and summary.

U3: Data Cleaning and Descriptive Analytics using Python

Teaching Hours: (6)

Data preprocessing and cleaning – Handling missing values – Data type conversion – Sorting and filtering data – Measures of central tendency using Python – Measures of dispersion – Measures of shape – Data summarization using Pandas – Basic exploratory data analysis (EDA).

U4: Data Visualization and Inferential Analytics

Teaching Hours: (6)

Introduction to data visualization – Importance of visual analytics – Visualization using Matplotlib and Seaborn – Bar charts, line charts, histograms, box plots, scatter plots – Introduction to inferential statistics – Hypothesis testing using Python – t-test and ANOVA (introductory level) – Correlation and simple linear regression.

U5: Applied Data Analytics and Case Studies

Teaching Hours: (6)

End-to-end data analytics process – Dataset selection and problem definition – Data preparation and analysis using Python – Visualization and interpretation of results – Mini case studies using real-world datasets – Business interpretation and reporting of insights.

Essential Reading:

Wes McKinney. (2017). *Python for Data Analysis*. O'Reilly Media

Grus, J. (2019). *Data science from scratch: first principles with python*. O'Reilly Media.

Anderson, D. R., Sweeney, D. J., Williams, T. A., Freeman, J., & Shoesmith, E. (Latest Ed). *Statistics for Business and Economics*. Cengage Learning, Inc..